# Data Source Discovery and Integration Architecture for "Fessi": A Knowledge Graph for Intelligent E-Waste Management

## Executive Summary

The transition to a circular economy relies fundamentally on the ability to decouple economic activity from the consumption of finite resources. For the waste management sector, specifically within the context of a large German university, this challenge manifests as a data problem. To divert Consumer Electronics (CE) and Waste Electrical and Electronic Equipment (WEEE) from landfill to high-value recycling or reuse pathways, the disposal decision—often made in seconds by a student or staff member—must be informed by deep regulatory, technical, and material insights. This report outlines the data architecture for "Fessi," a knowledge graph-based chatbot designed to bridge the gap between a physical object and its legally compliant disposal route.

The research conducted for this report reveals a fragmented but data-rich ecosystem. No single "master database" exists that maps a barcode directly to a specific bin in a specific building while accounting for German environmental law. Instead, the "Fessi" knowledge graph must function as a federation engine, synthesizing data from three distinct domains: **Regulatory Registries**, **Product Attribute Repositories**, and **Circular Economy Intelligence**.

Our exhaustive analysis identifies **Stiftung EAR (Elektro-Altgeräte Register)** as the foundational source for the "Logic & Rules" layer of the graph.[1] As the clearinghouse for the German *ElektroG*, EAR provides the definitive taxonomy for device categorization (the "Sammelgruppen"). However, regulatory data lacks the granularity required for user-centric identification. To resolve this, we have identified **Icecat** [2] and **EPREL** [3] as critical sources for the "Core Entities" layer, providing the technical specifications—such as screen dimensions and battery chemistry—necessary to trigger the conditional logic of disposal.

Crucially, the determination of hazardous waste status (AVV codes) requires a deep understanding of material composition. The report highlights the **ECHA SCIP Database** [4] and **iFixit API** [5] as essential sources for identifying Substances of Very High Concern (SVHC) and component separability. By integrating these sources, Fessi can infer that a specific laptop contains a lithium-ion battery requiring separation (BattG compliance) or that a legacy monitor contains mercury backlighting (hazardous waste).

This document serves as the comprehensive technical specification for ingesting, mapping, and harmonizing these data sources. It details the API endpoints, data models, licensing constraints, and semantic integration strategies required to build a compliant, robust, and intelligent waste management assistant for the university environment.

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# 1. The Regulatory and Data Landscape

The design of the "Fessi" knowledge graph cannot be separated from the legal framework it serves. In Germany, the disposal of electronic waste is governed by a strict hierarchy of European Directives and national laws. The data sources selected for this project are valuable precisely because they map to specific obligations within this hierarchy. Understanding these obligations is the prerequisite for defining the knowledge graph's schema.

## 1.1 The WEEE Directive and ElektroG: The Categorization Challenge

The overarching framework is the EU Directive 2012/19/EU on Waste Electrical and Electronic Equipment (WEEE), transposed into German law as the *Elektro- und Elektronikgerätegesetz* (ElektroG). The core challenge for Fessi is that the legal categorization of a device determines its disposal pathway, yet these categories are often opaque to the end-user.

The *ElektroG* mandates the separation of WEEE into distinct "Sammelgruppen" (collection groups). For a university chatbot, accurately classifying a device into one of these groups is the primary success metric. The groups are roughly defined by physical characteristics and handling requirements:

- **Group 1**: Heat exchangers (fridges, AC units).
- **Group 2**: Screens and monitors with a surface area greater than 100 $cm^2$.
- **Group 3**: Lamps.
- **Group 4**: Large equipment (any external dimension > 50 cm).
- **Group 5**: Small equipment (no external dimension > 50 cm).
- **Group 6**: Small IT and telecommunications equipment (no external dimension > 50 cm).

The data implication here is immediate: Fessi needs authoritative data on **dimensions** (to distinguish Group 4 from Group 5) and **device type** (to distinguish Group 5 from Group 6). A generic "product name" lookup is insufficient. A "printer" could be Group 4 (large office multi-function unit) or Group 5 (small home photo printer). The distinction dictates whether it fits in a campus "Red Bin" or requires a pickup order.

## 1.2 The Battery Act (BattG): The Chemistry Challenge

The *Batteriegesetz* (BattG) introduces a second, orthogonal layer of complexity. It mandates that batteries must be removed from WEEE prior to disposal unless they are enclosed in the device in a way that prevents ready removal by the end-user. Furthermore, batteries

themselves are classified into three types:

1. **Gerätebatterien (Portable)**: Standard consumer batteries.
2. **Industriebatterien (Industrial)**: Batteries designed for exclusively industrial use or electric vehicle propulsion.
3. **Fahrzeugbatterien (Automotive)**: Starter batteries.

For Fessi, this creates a critical data requirement: **Battery Chemistry and Removability**. The user must be instructed to "Remove the battery" if possible. If the battery is Lithium-Ion, specific safety warnings regarding fire risk and taping terminals are required.[6] If the battery is an E-Bike battery (common on campuses), it falls under "Industrial" and cannot be placed in the standard retail collection boxes (green boxes) found in university hallways. The graph needs to know *what* battery is inside the device.

### 1.3 The AVV Waste Catalogue: The Hazardous Challenge

Finally, the disposal process culminates in the assignment of a waste code from the *Abfallverzeichnis-Verordnung* (AVV). This is crucial for the university's facility management, which must document the disposal of hazardous waste. The relevant codes for electronics often depend on the presence of hazardous components:

- **16 02 13***: Hazardous equipment containing hazardous components (e.g., mercury switches, glass from cathode ray tubes).
- **16 02 14**: Discarded equipment other than those mentioned in 16 02 09 to 16 02 13.
- **20 01 35***: Discarded electrical and electronic equipment other than those mentioned in 20 01 21 and 20 01 23 containing hazardous components.

The asterisk (*) denotes hazardous waste. The knowledge graph must therefore contain logic to infer hazardous status. Does the "Old Projector" contain a mercury arc lamp? If yes, it is 20 01 35*. If it is a modern LED projector, it might be non-hazardous. This inference requires data on **Components** and **Materials**, which is rarely found in standard marketing APIs.

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## 2. Core Entity Data Sources

The "Core Entities" layer of the Fessi graph represents the "Things" in the world: Products, Brands, and Categories. The goal is to maximize coverage of the devices likely to be found on a university campus (laptops, monitors, lab equipment, smartphones) and link them to technical specifications.

### 2.1 Icecat Open Catalog

Icecat is a dominant player in the syndication of product content for e-commerce. Its "Open Icecat" project provides a massive, freely accessible repository of product datasheets, making

it a cornerstone for the Fessi graph.

### Source Identification

- **Name**: Open Icecat
- **URL**: https://icecat.biz/
- **Type**: Commercial Product Information Management (PIM) syndicate with an open data tier.
- **Documentation**: https://iceclog.com/manual-for-icecat-json-product-requests/ [7]

### Data Format & Access

Icecat offers two primary mechanisms for data ingestion, both relevant to Fessi [2]:

1. **XML/JSON Repository Download**: A bulk download of the entire catalog. This is massive (hundreds of gigabytes) but allows for offline processing and indexing of relevant categories.
2. **Real-time Live API**: A RESTful interface where a request is made by GTIN (EAN/UPC) or Brand + Product Code.
    - *Endpoint Structure*: https://live.icecat.biz/api?lang=<LANG>&ProductCode=<CODE>&Brand=<Brand>.[7]
    - *Response Format*: JSON or XML containing nested feature groups.

The XML schema is highly structured. It defines products not just as blobs of text, but as collections of "Feature Groups" (e.g., "Memory", "Display", "Battery"). This structure is ideal for mapping to a knowledge graph ontology.

### Coverage & Quality

- **Coverage**: Icecat is exceptionally strong in **Information Technology (IT)** and **Consumer Electronics (CE)**. It covers virtually every major brand of laptop, monitor, smartphone, printer, and peripheral likely to be used by students and faculty.
- **Quality**: The data is "manufacturer authorized," meaning it comes directly from brands like HP, Dell, and Samsung. This ensures high accuracy for technical specs like dimensions and weight.
- **Limitations**: Coverage of obscure "white label" devices or specialized laboratory equipment (e.g., centrifuges, microscopes) is limited.

### Relevance Assessment

Icecat is the primary source for **Physical Attributes** required by ElektroG.

- **Dimensions**: The standard schema includes Height, Width, Depth. This allows Fessi to deterministically calculate if a device is > 50 cm (Category 4 vs. Category 5).
- **Weight**: Essential for the university's waste balance reporting (waste is measured in metric tons).
- **Battery Details**: Icecat datasheets frequently list battery technology (e.g., "Lithium-Ion"),

cell count, and sometimes weight.[9] This is critical for BattG compliance.
- **Connections**: Data on ports (e.g., "USB 2.0", "VGA") can help infer the age of a device if the release date is missing, which serves as a proxy for identifying legacy hazardous components (e.g., VGA monitors are more likely to be CCFL/Mercury based).

### Integration Notes

For the Fessi graph, we recommend a hybrid approach. The core graph should be pre-populated with "Brand" and "Category" nodes from Icecat's reference files (Refs file). When a user scans a barcode, the system should query the Icecat Live API.

- **Mapping**:
  - Icecat Feature Battery technology -> Fessi Property hasBatteryChemistry.
  - Icecat Feature Display diagonal -> Fessi Property screenSize.

## 2.2 EPREL (European Product Registry for Energy Labelling)

The European Product Registry for Energy Labelling (EPREL) is an EU-managed database. Since 2019, suppliers have been legally required to register products that require an energy label before placing them on the EU market.

### Source Identification

- **Name**: EPREL (European Product Registry for Energy Labelling)
- **URL**: https://eprel.ec.europa.eu/ [3]
- **Type**: Regulatory Database.

### Data Format & Access

- **Public Access**: Consumers can browse the website or scan QR codes on energy labels.
- **API Access**: A public API exists but has strict access controls for bulk data. Accessing the full machine-readable dataset requires an API key obtained via a digitally signed "EPREL Declaration" using a Qualified Electronic Seal (QES).[3]
- **Data Model**: The data is structured according to product-specific regulations (delegated acts). A "Refrigerating Appliance" has a different schema than an "Electronic Display."

### Coverage & Quality

- **Coverage**: High for specific regulated categories: **Electronic Displays (Monitors, TVs)**, **Lighting Sources**, **Refrigerators**, **Washing Machines**, **Dishwashers**.
- **Gap**: It does *not* cover laptops, smartphones (until recently/future regulation), or small kitchen appliances like toasters.
- **Quality**: This is regulatory-grade data. The dimensions and screen sizes are legally binding declarations, making them more trustworthy than third-party marketing data.

### Relevance Assessment

EPREL is the "Truth Source" for **Category 2 (Screens)** and **Category 3 (Lamps)**.

- **Screen Area**: ElektroG distinguishes screens based on a surface area of 100 $cm^2$. EPREL data explicitly contains visibleScreenArea (in $dm^2$ or $cm^2$), allowing for a mathematically precise classification that avoids user estimation errors.[11]
- **Lighting Technology**: EPREL classifies light sources (e.g., "LED", "OLED", "Compact Fluorescent"). This is the single most important data point for separating hazardous gas-discharge lamps (Mercury) from non-hazardous LED bulbs.

### Integration Notes

Given the strict API requirements (QES), the university should apply for an API key as a "Public Institution" or "Researcher."

- **Workflow**:
  1. User scans QR code on energy label (if present).
  2. Fessi extracts the EPREL Model Identifier from the URL.
  3. Fessi queries EPREL API for lightSourceTechnology or panelTechnology.
  4. **Logic Trigger**: IF technology == "Fluorescent" THEN Instruction = "Hazardous! Do not break. Use container X."

## 2.3 Open Products Facts (OPF)

While Icecat is commercial and EPREL is regulatory, Open Products Facts (a sibling to Open Food Facts) is a crowdsourced, open-data initiative.

### Source Identification

- **Name**: Open Products Facts
- **URL**: https://world.openproductsfacts.org/ [13]
- **Type**: Open Data Community Project.

### Data Format & Access

- **API**: A robust JSON API (https://world.openproductsfacts.org/api/v2/product/[barcode].json).
- **Exports**: Daily CSV and MongoDB dumps are available.[13]
- **Taxonomy**: The system uses a hierarchical taxonomy for categories, labels, and brands, managed via a taxonomy editor on GitHub.[14]

### Coverage & Quality

- **Coverage**: Growing but sparse compared to food. Coverage of electronics is spotty but includes unique data points like packaging material (e.g., "Cardboard box," "Plastic film") which are useful for advising on packaging waste disposal alongside the device.
- **Quality**: Crowdsourced, so verification is needed.

### Relevance Assessment

OPF is valuable as a **fallback source** for obscure items and for its unique focus on **Packaging**.

- **Packaging Waste**: Fessi must advise not just on the device, but the box it came in. OPF often lists packaging hierarchies (e.g., packaging: box -> paper).
- **Eco-Score**: The platform is experimenting with "Eco-Scores" and repairability indices, which can enrich the "sustainability advice" portion of the chatbot.

### Integration Notes

- Use the ecoscore and packaging fields to provide holistic disposal advice (e.g., "Dispose of the phone in the Red Bin, but put the box in the Paper Blue Bin").

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# 3. Regulatory Logic Sources: The "Rules" Layer

The "Core Entities" layer provides the "What." The "Rules" layer provides the "How." This data comes from government registers and technical guidelines.

## 3.1 Stiftung EAR: The Classification Authority

The **Stiftung EAR** data is not just a list of names; it is the taxonomy of the German waste system.

### Detailed Source Profile

- **Dataset**: List of Registered Producers and Authorized Representatives.[1]
- **Data Fields**:
  - Hersteller Name (Manufacturer)
  - Marke (Brand)
  - Geräteart (Device Type) - e.g., "Kleingeräte, die in privaten Haushalten genutzt werden können."
  - WEEE-Reg.-Nr.
  - Marktaustrittsdatum (Market Exit Date)

### Relevance Assessment & Integration

This dataset allows Fessi to validate if a device is "Orphan Waste" (produced by a manufacturer that no longer exists or isn't registered).

- **B2B vs B2C Logic**: The EAR register explicitly flags brands/categories as **B2B (Business-to-Business)** or **B2C (Business-to-Consumer)**.
  - *Insight*: A university environment contains both. A student's "Sony Headphones" are B2C. A lab's "Network Switch" might be registered as B2B.

    - *Graph Logic*: When a user identifies a "Cisco Switch," Fessi checks the EAR register for Cisco's registration in that category. If it is "Only B2B," the chatbot **must not** direct the user to the public collection bins (which are for B2C waste). Instead, it must direct them to the university's asset disposal department. This prevents the university from illegally disposing of commercial assets in municipal streams.

### Facility Mapping (Infrastructure Layer)

The EAR also publishes the list of **First Treatment Facilities (EBA)**.[1]

- **Usage**: While the chatbot primarily directs to campus bins, this list provides the valid "End Nodes" for the waste graph. It ensures that the university's internal logistics eventually route to certified partners.

## 3.2 LAGA Mitteilung 31: The Logic of Waste Codes

The **LAGA Mitteilung 31** is a technical guideline that acts as the "translation manual" between physical components and **AVV Waste Codes**. This is the most complex logic Fessi must implement.

### Detailed Analysis of Classification Logic

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The guideline uses conditional logic based on the state of the device.

| Component / State | Condition | Waste Code (AVV) | Graph Inference Rule |
|---|---|---|---|
| **Complete Device** | Small IT (e.g., Mouse, Keyboard) | 20 01 36 | IF Category=6 AND Hazardous=False -> 20 01 36 |
| **Complete Device** | Contains Asbestos (e.g., old heater) | 20 01 35* | IF Material=Asbestos -> Hazardous |
| **Component: PCB** | Standard (Green board) | 16 02 16 | IF Part=PCB -> Check(Components) |

| | | | |
|---|---|---|---|
| **Component: PCB** | Contains Capacitor > 25mm height/diameter | 16 02 15* | IF Part=PCB AND HasCapacitor(>25mm) -> Hazardous |
| **Component: Cable** | External standard cable | 17 04 11 | IF Part=Cable -> Non-Hazardous |
| **Screen (CRT)** | Cathode Ray Tube | 20 01 21* | IF Tech=CRT -> Hazardous |

### Integration Recommendations

Fessi cannot "download" LAGA 31 as a database. It must be **encoded** as a set of SHACL rules or SWRL (Semantic Web Rule Language) logic within the graph.

- **Implementation**: Create a "HazardCheck" routine. When a device is identified (e.g., via Icecat), the system checks its attributes against the LAGA rules.
- **User Interaction**: If data is missing (e.g., Icecat doesn't say if the capacitor is >25mm), the chatbot should default to the "Worst Case" (Hazardous) or prompt the user: "Is the device physically damaged?" (Damaged lithium batteries change the disposal pathway entirely).

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# 4. Material & Circularity Intelligence Sources

To apply the LAGA rules, Fessi needs to know what is *inside* the box. This requires "Teardown Intelligence."

## 4.1 iFixit API: The Component Graph

iFixit is more than a repair site; it is a database of device composition.

### Source Identification

- **Name**: iFixit API
- **URL**: https://www.ifixit.com/api/2.0/doc [5]
- **Endpoints**: /teardowns, /guides/{id}, /parts.

### Data Format & Access

The API returns JSON objects detailing the hierarchy of a device.

- **parts Object** [17]: An array of components found in the device.
  - *Example*: {"text": "iPhone 6 Battery", "type": "part"}.

- **tools Object** [17]: An array of tools required to open the device.

### Relevance Assessment

iFixit data serves as a proxy for **Separability** and **Hazard Presence**.

1. **Battery Removability (BattG)**: The BattG requires users to remove batteries if possible. iFixit guides provide the answer.
    - *Logic*: IF a guide exists titled "Replacing the Battery" AND the difficulty is "Low" or "Moderate", THEN Fessi advises: "Please remove the battery and dispose of it in the Green Battery Box."
    - *Logic*: IF difficulty is "Very Hard" or tools includes "Heat Gun" (implying glued battery), THEN Fessi advises: "Do not attempt to remove battery. Dispose of the whole unit in the Red Bin."
2. **Material Identification**: The parts list often contains specific chemical identifiers in the names, such as "Mercury switch" or "Ni-Cd Battery".

### Integration Notes

- **Ingestion**: Map the iFixit Category hierarchy to the Fessi Category ontology.
- **Inference**: Use the existence of specific parts to set boolean flags on the Product Node: hasRemovableBattery, containsGlass, requiresSpecialTools.

## 4.2 ECHA SCIP Database: The Toxicology Layer

The **SCIP database** is the EU's central repository for information on **Substances of Concern In articles as such or in complex objects (Products)**.

### Source Identification

- **Name**: SCIP Database
- **Owner**: European Chemicals Agency (ECHA)
- **URL**: https://echa.europa.eu/scip-database [4]

### Data Format & Access

- **Format**: IUCLID (XML).
- **Access**: Dissemination portal and full database export.
- **Structure**: The data links **Article Categories** (CN Codes) to **Candidate List Substances** (SVHCs).

### Relevance Assessment

This is the only source that links a product to specific chemical hazards mandated by **REACH**.

- **Example**: Identifying that a specific "PVC Cable" contains "Lead stabilizers" (>0.1% w/w).
- **Impact**: While the user doesn't need to know the chemical name, Fessi uses this to determine if the item belongs in general e-waste or a specific hazardous waste flow

(Sondermüll).

### Integration Notes

The full SCIP export is massive and complex (hierarchical XML).

- **Strategy**: Do not attempt real-time queries. Instead, extract a "Risk Map" for the Consumer Electronics categories.
- **Mapping**: Map Candidate List Substance to AVV Hazard Trigger. If a product category frequently contains "Lead" or "Cadmium" in SCIP, increase the "Hazard Probability" score in Fessi.

## 4.3 Open Repair Alliance (ORDS)

The **Open Repair Data Standard (ORDS)** aggregates data from community repair events (Repair Cafés).[19]

### Data Format & Access

- **Format**: CSV files following the ORDS schema.
- **Fields**: Product Category, Brand, Problem, Repair Outcome, Repair Barrier.

### Relevance Assessment

This data provides **statistical probabilities** of failure modes.

- **Insight**: If ORDS data shows that 80% of "Kettles" from "Brand X" fail due to "Scale buildup" (repairable) rather than "Heater failure" (fatal), Fessi can suggest "Descaling" before offering disposal instructions. This aligns with the "Reuse" priority of the circular economy hierarchy.

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# 5. Battery Disposal Specifics (BattG)

Batteries represent the highest risk flow in university waste. Mixing lithium batteries with paper or general waste causes fires.

## 5.1 EPBA & Battery Chemistry Identification

The **European Portable Battery Association (EPBA)** provides guidelines on battery marking.[6]

- **Chemistry Codes**: Li-ion (Lithium Ion), Ni-MH (Nickel Metal Hydride), Ni-Cd (Nickel Cadmium), Pb (Lead Acid).
- **Visual Symbols**: The "Crossed-out Wheeled Bin" with chemical symbols (Cd, Hg, Pb) beneath it.

### 5.2 Regulatory Distinctions (EAR & BattG)

Fessi must strictly implement the legal distinction between battery types [1]:

1. **Portable (Gerätebatterien)**: < 5kg, sealed. -> **Destination**: Green Collection Boxes (Retail/Campus).
2. **Industrial (Industriebatterien)**: This definition includes **all batteries for electric vehicles**, including **E-Bikes** and **E-Scooters**.
    - *Critical Logic*: Students often confuse E-Bike batteries with large portable batteries. Fessi must explicitly intercept this: "Is this an E-Bike battery?" -> "Do NOT put in the Green Box. Return to distributor or Facility Management."

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## 6. Emerging Standards: The Digital Product Passport

The future of waste data lies in the **Digital Product Passport (DPP)** and **GS1 Digital Link**.

### 6.1 GS1 Digital Link

This standard replaces the traditional barcode with a web-enabled QR code.[23]

- **Structure**: https://id.gs1.org/01/{GTIN}/21/{Serial}.
- **Relevance**: It allows Fessi to access manufacturer-hosted data directly. The **GDSN** (Global Data Synchronization Network) standard includes attributes for hazardousWasteCode and dangerousGoods.[24]
- **Integration**: Fessi should be programmed to parse these URIs. If a student scans a QR code, Fessi can extract the GTIN and query the GDSN resolver for safety data sheets.

### 6.2 TCO Certified

**TCO Certified** is a sustainability certification for IT products.[26]

- **API**: They offer an API to check if a GTIN is certified.
- **Relevance**: Certified products meet strict criteria for hazardous substances and battery replaceability. A positive hit on the TCO API is a strong signal that the device is "Cleaner" and easier to disassemble.

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## 7. Integration Architecture: The "Fessi" Pipeline

To make this actionable, we propose the following data pipeline for the Knowledge Graph.

### 7.1 Data Ingestion Layers

1. **The Skeleton (Ontology)**: Built manually from **EAR** (Categories) and **LAGA 31** (Rules).

    - *Node*: Category:Screens. *Rule*: IF area > 100cm².
2. **The Population (Batch)**:
    - Import **EAR Producer List** CSV to populate valid Brands.
    - Import **Icecat** XML reference files to populate common Product Models.
3. **The Enrichment (Real-Time)**:
    - **User Query**: "Dispose of Samsung Galaxy S21".
    - **Step 1**: Lookup Product in Graph. If missing, query **Icecat API** / **Open Products Facts**.
    - **Step 2 (Attributes)**: Retrieve Battery: Li-Ion, Weight: 170g.
    - **Step 3 (Logic)**:
        - Check **BattG Rule**: "Is battery removable?" (Query **iFixit** or check Icecat removable flag).
        - Result: "Integrated Battery."
    - **Step 4 (Classification)**:
        - EAR Category: "Small IT" (Group 6).
        - Hazard Check: "Lithium Ion" = Fire Risk.
    - **Step 5 (Destination)**:
        - Map to University Infrastructure: "Group 6 Container (Red Bin) - Handle with Care."

### 7.2 Handling Gaps: The "Human-in-the-Loop"

Data will often be missing.

- **Gap**: No dimensions available for an old printer.
- **Strategy**: Fessi must ask the user: "Is the printer larger than a standard backpack (50cm)?"
- **Gap**: Unrecognized Brand.
- **Strategy**: Default to "Hazardous/Caution" workflow and direct to a staffed collection point (University Recycling Center) rather than an unmonitored bin.

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## 8. Conclusion

The "Fessi" project has the potential to transform waste management from a compliance burden into an intelligent, data-driven service. By anchoring the knowledge graph in the legal authority of **Stiftung EAR**, enriching it with the technical depth of **Icecat** and **EPREL**, and refining it with the circular intelligence of **iFixit** and **SCIP**, the university can ensure high-quality separation of waste. This not only ensures compliance with German law but actively recovers valuable resources, fulfilling the promise of the circular economy. The blueprint provided in this report offers the comprehensive data foundation required to build this system.

**Works cited**


1. Lists | stiftung ear | stiftung elektro-altgeräte register, accessed January 10, 2026, https://www.stiftung-ear.de/en/lists/
2. Icecat: AI Platform for Content Management & Syndication, accessed January 10, 2026, https://icecat.com/
3. Request Public API Key - EPREL - European Union, accessed January 10, 2026, https://eprel.ec.europa.eu/screen/requestpublicapikey
4. ECHA SCIP Database Available for General Public - APA Engineering, accessed January 10, 2026, https://apaengineering.com/compliance-blog/echa-scip-database/
5. API v2 endpoints - Knowledge Base - i-doit, accessed January 10, 2026, https://kb.i-doit.com/en/i-doit-add-ons/api/methods/v2/index.html
6. European Portable Battery Association Position - EPBA, accessed January 10, 2026, https://www.epbaeurope.net/assets/EPBA-Position-Batteries-Regulation-Proposal-final-(1).pdf
7. Manual for Icecat JSON Product Requests - Iceclog, accessed January 10, 2026, https://iceclog.com/manual-for-icecat-json-product-requests/
8. What is a good model, using Open IceCat to import data and populate products in a a catalog's database - Stack Overflow, accessed January 10, 2026, https://stackoverflow.com/questions/8298049/what-is-a-good-model-using-open-icecat-to-import-data-and-populate-products-in
9. Integrated Dell Remote Access Controller 9 User's Guide - Icecat, accessed January 10, 2026, https://objects.icecat.biz/objects/mmo_99303292_1681068469_3758_12306.pdf
10. EPREL Application Programming Interface ('API') to access registered products public data TERMS AND CONDITIONS - European Commission, accessed January 10, 2026, https://ec.europa.eu/assets/move-ener/eprel/EPREL%20Public/Public%20API%20Term%20and%20Conditions/API_TERMS_AND_CONDITIONS_EN.pdf
11. The Product Database EPREL - LightingEurope, accessed January 10, 2026, https://www.lightingeurope.org/images/2024-10-25-Webinar-Verification-pt.pdf
12. EPREL - European Commission - Energy Efficient Products, accessed January 10, 2026, https://energy-efficient-products.ec.europa.eu/eprel_en
13. Data, API and SDKs - Open Products Facts, accessed January 10, 2026, https://world.openproductsfacts.org/data
14. Taxonomies are at the heart of Open Food Facts data structure - this project provides an editor - GitHub, accessed January 10, 2026, https://github.com/openfoodfacts/taxonomy-editor
15. Anforderungen zur Entsorgung von Elektro- und Elektronik-Altgeräten, accessed January 10, 2026, https://www.laga-online.de/documents/m31_merkblatt_elektroaltgeraete_1503993087.pdf
16. Documentation - oManual: The Open Manual Format, accessed January 10, 2026,

https://www.omanual.org/documentation.php
17. API v2.0 - Wikis Documentation - Rat Rig, accessed January 10, 2026, https://ratrig.dozuki.com/api/2.0/doc/Wikis
18. Get Business Ready for ECHA SCIP Database | APA Engineering, accessed January 10, 2026, https://apaengineering.com/chemical-regulatory-compliance/service/scip-database
19. Open Repair Alliance Datasets 2021 • repairData, accessed January 10, 2026, https://petzi53.github.io/repairData/
20. Open Repair Data Standard, accessed January 10, 2026, https://standard.openrepair.org/
21. Guidelines on Portable Batteries and Battery Peripherals Marking Requirements in the European Union - EPBA, accessed January 10, 2026, https://www.epbaeurope.net/assets/New-battery-marking-guidelines-January-2015.pdf
22. Your authority for implementing ElektroG & BattG | stiftung elektro-altgeräte register, accessed January 10, 2026, https://www.stiftung-ear.de/en/
23. GS1 Digital Signatures Technical Implementation Guideline, accessed January 10, 2026, https://www.gs1.org/standards/gs1-digital-signatures/current-standard
24. GS1 Attribute Business Definitions Standard, accessed January 10, 2026, https://www.gs1.org/docs/gdsn/GS1_Attribute_Business_Definitions.pdf
25. GS1 US Guidance for Sharing Product Attributes via GDSN in Retail Grocery, accessed January 10, 2026, https://documents.gs1us.org/adobe/assets/deliver/urn:aaid:aem:99b5d28c-1bbe-4858-bfe6-8a59e0de9b05/Guideline-GS1-US-Guidance-for-Sharing-Product-Attributes-via-GDSN-in-Retail-Grocery.pdf
26. Join our pilot for displaying certified products in reseller webshops, accessed January 10, 2026, https://industry.tcocertified.com/gtin-api/